[illegible]

(A)

[illegible] LECTURE NOTES (BIOLOGY [illegible])

Course outline

1. Microscope
2. Microbial Contamination
3. Heating Apparatus (a) Autoclave (b) Incubator, (c) Centrifuge and Water bath.
4. Microtome
5. Artificial Respiration.

Course Lecturer [illegible]

1. Microscope

This is a scientific instrument used to view [illegible] organisms or things that cannot be seen with [illegible] eye (naked eyes).

micro - small organisms or things

scope - view hence the name microscope.

Types of Microscope

a) Compound microscope (light microscope)
b) Electron microscope
c) Phase Contrast microscope
d) Interference microscope
e Florescence microscope
f Polarizing microscope
g Dark field microscope.

The mostly commonly used is compound microscope. This consist of two converging lenses which are seperated by distance which is known as eye piece lens. Magnification of compound microscope is objective lens X Eye piece lens.

PARTS OF MICROSCOPE AND FUNCTIONS

1. Eye piece: This is lens system near the eye which magnifies the primary image of the object so as to form a virtual image away from the eye point. With magnification stamp of X10

2 Rotatable head: To move the eyepiece to the suitable position.

3 Arm: To carry the instrument comfortably.

4 Objective lenses: This is lens system which magnify the image of the object to form the primary image. They are four in number with magnification on each of them: X4, X10, X40 and X100 oil immersion.

5 Condenser: This is optical element that [illegible] the light rays and ~~collects~~ to converge it [illegible] the stage through the small opening ~~for~~ to a focus.

6 Base: This is the rectangular part that houses the Illuminator, it is used for balancing on a smooth surface.

7 Illuminator: This is to provides light either from sunlight or electricity and is housed at the base of the microscope.

8 Iris Diaphragm: This is mechanical device mounted within the condenser, it controlled the amount of light rays enter into objective lens for in order to achieve optimum objective performance.

9 Nose Piece: This is the revolver that carries the objectives, it is hold to turn to the desired objective lens to be use.

10 Coarse Adjustment knob: This is knob using in moving the stage clockwise and anti-clockwise direction to achieve a focus. It is seen by the side of the microscope.

11 Fine Adjustment knob: This is commonly found at the same place with coarse Adjustment knob and is to give a fine and clearer focus.

CARE AND MAINTANCE OF A MICROSCOPE

1 The mechanical and optical system must be handled with care.

2 Always carry the microscope with two hand

right at the arm and left hand under the base.
3 Keep the microscope on a flat surface.
4 Wash all the microscope slide with soap and water after using (This should be neutral detergent)
(5) Do not use organic solution to wipe the surface of various components.
6 Never detach the eye piece after use. Lower driving adjustable knob use to
7 No piece will be seen while using objective
8 Always keep the microscope inside the bag or box after using for safety.
9 Never disassemble the microscope for repair but reserve the task strictly for experienced technician.

MICROBIAL CONTAMINATION

Ever science the development of techniques to obtain micro organisms in pure culture, the susceptibility of such cultures to the unwanted growth of other microbes has been recognized.
This can be as a result of microbial contamination of culture media.
In morden medicine and science, the importance of hand washing and the maintenance of a sterile operating theatre is taken for granted.

PREVENTION OF MICROBIOLOGICAL CONTAMINATION BEGINS IN THE LABORATORY.

A. Variety of preventive procedures are a common part of an efficient microbiology Laboratory. The use of sterile equipments and receptacles for Liquid and solid growth media is a must. The prevention of contamination during the manipulations of microorganisms in the laboratory falls under the term "aseptic technique".
Examples of aseptic technique include the disinfection

of work surfaces and the hands of the relevant lab personnel before and after contact with microorganisms and the flaming of the metal loops or rod used to transfer bacteria from one location to another.

FIRST AID MEASURE FOR MICROBIAL CONTAMINATION

In case of contact, wash contaminated skin thoroughly with antiseptic soap and water.

If wound contamination is suspected, seek immediate medical attention. In case of ingestion/inhalation, seek immediate medical attention, ... [illegible] the name of micro organism.

MICROTOME

Microtome is a machine used for cutting thin section from piece of tissue. This machine is used in study of histology which is the branch of medical science concern with examination of tissue. It is made up of following components:

COMPONENTS OF MICROTOME

1. A knife held in a firm support.
2. A device for holding the piece of tissue firmly.
3. A mechanism for advancing the piece of tissue across the knife blade or the knife across tissues.
4. A mechanism for advancing the piece of tissue across very small and accurately measure distance so that section of suitable thickness can be obtained.

TYPES OF MICROTOME

1. Cambridge Rocker Microtome.
2. Sledge Microtome with moving block.
3. Sledge Microtome with moving knife.
4. Rotary Microtome.
5. Freezing Microtome.

STANDARD PROCEDURES FOR CARE OF [illegible] INJURIES

1. General Instructions:

A. Provide total care for those injuries clearly within your capability to handle. All questionable case should be referred to a local doctor or to the emergency room.

B. Wear disposable gloves and wash your hands thoroughly before and after treating an injury.

2. CARE OF INJURIES

A. Minor laceration (Cuts): Ge[illegible] [illegible] the wound may apply antibiotic ointment. Cover with appropriate dressing.

B. Severe Laceration (cuts): If fresh bleeding, apply pressure dressing and refer to doctor.

C. Minor Cutusions (bruses) Apply c[illegible] compresses. If skin is broken, gently cleanse affected area. Antibiotic may be apply, cover with appropriate dressing.

D. Minor abrasions (Scrapes: Gently cleanse area. Antibiotic may be applied and cover with appropriate dressing.

e. Minnor burns: Immerse burned area in cold water for 10-15 minutes, or until pa[illegible] is relieved. Gently cleanse area, apply moist dressings and bandage loosely.

f. Foreign body in eye: If foreign body cannot be easily removed by touching it with a cotton tipped swab or by irrigating it with eye wash, refer to doctor.

CONTENTS OF THE FIRST AID BOX

- ✓ Name and phone number also address of the first Aiders or appointed person.
- ✓ Guidance leaflet.
- ✓ 20 individually wrapped st[illegible]

dressings (assorted sizes) appropriate to the type of work
✓ 2 Sterile eye pads with attachment
✓ 4 individually wrapped triangular bandages.
✓ 6 Safety pins.
✓ 6 medium sized individually wrapped sterile Un-Medicated wound dressings (12cm X 12cm)
✓ 2 medium sized individually wrapped sterile Un-medicated wound dressings (18cm x 18cm)
✓ One pair of disposable gloves
✓ Clinical waste bag & instruction for use

✓ LABORATORY SAFETY RULES

There are certain basic rules to be followed for Safe working in the Laboratories;

* Wear safety glasses and a fastened Lab-Coat at all times.
> Wear gloves if appropriate.
> Do not eat or drink or take food into the Laboratory.
> Keep long hair tied back.
> Do not use a personal Stereo or any other equipment involving headphone.
> Use all chemicals carefully and in a fume hood whenever possible.
> Dispose of Solvents properly.
> Beware of glass especially if it is broken and then dispose of it Safely.
> Keep your work area tidy and clean up any spills including water on the floor.
> If emergency alarm sound, make your work Safe and leave the building as directed.
> Report any accident you have.
> Wash your hand thoroughly while leaving The Laboratory

(1) INTRODUCTION TO GENERAL LABORATORY TECHNIQUES. (GLT 111)
BIOLOGY OPTION. NDI 2024/2025 ACADEMIC SESSION.
LECTURE NOTE BY: MR. MICAH ORNAN.

INTRODUCTION:

In a scientific Investigation, there is an attitude that demands of one to explore his environment or to develope the relevant hypothesis in Order to establish its acceptability or otherwise. This type of Investigation can be carried out in a laboratory.

LABORATORY:

A Laboratory is a Setting or Place in Order to try out Specific information and Test principles and theories in Science.

It is thus, a place or room where Scientific experiments are carried out. It is appropriately equiped with Various items that are required to provide answers to introguing questions in science.

LABORATORY HAZARDS:

Working in a Laboratory is characterized by certain specific features which in-turn make certain demands on the laboratory users.

There is need for special attention in Order to adhere to necessary safety rules so as to minimize the occurance of accident. There is also need to ensure that all experiments, reactions and related operations are carried out accurately.

HAZARD:

Hazards can be defined as un-safe act that can bring about Injury. Such Injuries may be as a result of improper handling of glasswares, apparatus, reagents, chemicals and any other act of carelessness due to lack of obeying some elementary laboratory safety rules.

(2) SOME SAFETY LABORATORY RULES.

There are certain basic rules to be followed for Safety Working in the laboratory such as :—

1. Do not eat or drink or take any food in the laboratory.
2. Beware of glasses especially if it is broken.
3. Keep your work area tidy and clean up any spills of water on the floor.
4. Keep Long hair tied back and avoid using hill shoes.
5. Be Careful and Concentrate on your work.
6. Wear your Lab-Coat at all the time.
7. Wash your hands Carefully before you leave the Lab.
8. Report any accident you have to your supperior.

CONTENTS OF FIRST AID BOX.

As a first Aider, you Must have the following Contents in your box.

First aid is also defined as the first treatment taken to the patient before taking the Patient to the Hospital or Doctore if necessary.

1. Individual Wrapped sterile adhensive dressing appropriate to the type of work.
2. Individual Wrapped triangular bandage.
3. Safety Pins.
4. Un medicated wound dressing.
5. Pairs of disposible gloves and noseMask.
6. Clinical Waste bag.
7. Packets of Razor blade.
8. Water/cottonwool/Iodin Solution
9. Sterilizer etc.

(3) MICROSCOPE.

A Microscope is a scientific Instrument used to Magnify and observe some small objects that are too tiny to be seen with the naked eye.

It works by using lenses or electron beams to enlarge the image of the specimen. A microscope is widely used in Sciense, Medicine and Industry for research and diagnostic purposes.

There are two types of Microscope commonly used these are (1) Light/ Compound Microscope (2) Electron Microscope.

1. Light/ Compound Microscope:

This type of Microscope uses visible light and lenses to magnify objects, commonly used Hospital and Biology Lab.

2. Electron Microscope:

This type Microscope uses electron beams for much higher Magnification, alowing scientist to see fine structures like Viruses and cell Organelles.

PARTS OF MICROSCOPE.

A microscope has several important parts which can be grouped into (1) Optical (2) Mechanical (3) Illuminating Components.

1. OPTICAL COMPONENTS:

i. Eyepiece:- Is the lens you look through usually magnifies the image.

ii Objective lens:- These are Multiple lenses with different magnifications (eg ×4, ×10, ×40, ×100 etc).

iii Revolving Nosepiece:- It holds and rotates the objective lenses.

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2. MECHANICAL COMPONENTS:

i Body tube:- This Connects the eyepiece to the Objective lenses

ii Arm:- This Supports the body tube and Connects it to the base

iii Stage:- Is the Plate form where the specimen slide is Placed.

iv Stage clip:- It holds the slide in Place.

v. Coarse adjustment Knob:- It moves the stage up and down for rough focusing.

vi Fine adjustment Knob:- It allows Precise focusing or fine focusing.

vii Base:- Is the bottom Part that Support the Microscope.

3. ILLUMINATING COMPONENTS:

i Light Source:- It Provide light for Viewing the Specimen

ii Condenser Lens:- It focus light to the Specimen.

iii Diaphram:- It Controls the amount of light from passing through the specimen.

(5) CARE AND MAINTENANCE OF A MICROSCOPE:

Here are some important guidelines for care and maintenance of a microscope:

1. Handling and storage:

i. Always carry the microscope with two hands. One on the arm and one hand on the supporting base.

ii Store the microscope in a clean, dry place, preferably covered when not in used.

iii Keep the microscope on a stable vibration in free surface to prevent damage.

2. Cleaning:-

i Always use lens paper to clean the lenses, never use tissue or finger for cleaning.

ii Clean the lenses with 70% alcohol using a cotton wool

iii Wipe the stage or body with a clean, dry cloth after use.

3. Proper Use:

i. Always start with the lowest objective lens eg ×4 before switching to the higher objective lens.

ii Use the coarse adjustment knob only with the low power objective lens to avoid crashing the lens in to the slide, and later on fine adjustment knob.

iii Adjust the diaphram and light source properly to get a clear image without stracing your eyes.

4 Preventive Maintenance:

i Regularly check ~~the~~ and tighten any loose screws or knobs.

ii Ensure that the light source works properly.

iii Keep the microscope away from water and moisture.

Electron Microscope.

The electron Microscope Use's electrons to illuminate an object. Electrons have smaller wavelength than light, so they can resolve much smaller structures. Magnetic field are used to create lenses that direct and focus the electrons. Since electrons are easily scattered by air molecules, the interior of the electron microscope must be used at a very high vacuum.

Types of Electron Microscope.

There are two types of electron Microscope, these are: (1) Transmission electron Microscope.
(2). Scanning electron Microscope.

1. Transmission Electron Microscope :- (TEM).
It is a powerful type of Microscope that allows scientist to see ultra-thin slices of specimens in high details. Electron beam is directed on the object to be magnified. Some of the electrons are absorbed while others bounce off the specimen or pass them through. It can magnify objects up to 1,000,000 times.

2. Scanning Electron Microscope. (SEM).
It focused beam of electrons to scan the surface of a specimen. The electron beam moves over the entire sample to create a magnified image of the surface of the object. It scans electron beam on an image on to the screen of a Television. It can also magnify objects up to 100,000 times.

NOTE: GENERALLY THERE ARE SIX (6) TYPES OF MICROSCOPE.

1. Compound Microscope.
2. Electron microscope.
3. Phase Contrast Microscope.
4. Interference Microscope.
5. Polarising Microscope.
6. Dark-field Microscope.

MICROTOME.

Microtome is a Machine use for Cutting thin Structures or sections from Pieces of tissue. This Machine is use in the study of Histology which is the branch of Medical Science Concerned with examination of tissue.

Many different Varieties of Microtome are available, but all of them are Made up of the following principal Components.

1. A Knife held in a firm Support
2. A device for holding the piece of tissue firmly.
3. A Mechanism for advancing the piece of tissue across the Knife blade or the Knife across the tissue.
4. A mechanism for advancing the piece of tissue across a very small and accurately measured distance so that section of suitable thickness Can be obtained.

TYPES OF MICROTOME.

There are five (5) types of microtome, these are.

1. Cambridge Rocker Microtome.
2. Sledge Microtome with Moving Knife.
3. Sledge Microtome with Moving block.
4. Rotary Microtome.
5. Freezing Microtome.